



Comparative study on upconversion luminescence and temperature sensing of α - and β -NaYF₄:Yb³⁺/Er³⁺ nano-/micro-crystals derived from a microwave-assisted hydrothermal route

Lili Tong^a, Xiangping Li^{a,*}, Ruinian Hua^b, Xuejing Li^a, Hui Zheng^a, Jiashi Sun^a, Jinsu Zhang^a, Lihong Cheng^a, Baojiu Chen^{a,*}

^a Department of Physics, Dalian Maritime University, Dalian 116026, China

^b College of Life Science, Dalian Nationalities University, Dalian 116600, China

ARTICLE INFO

Article history:

Received 23 April 2015

Received in revised form

6 July 2015

Accepted 14 July 2015

Available online 21 July 2015

Keywords:

Phosphor

Upconversion luminescence

Temperature sensing

Microwave-assisted hydrothermal

ABSTRACT

Yb³⁺/Er³⁺ co-doped α - and β -phase NaYF₄ nano-/micro-crystals were prepared through a microwave-assisted hydrothermal route. The crystal structure and microscopic morphology of the samples were characterized by means of X-ray diffraction (XRD) and scanning electron microscopy (SEM). Frequency upconverted emissions from the two thermally coupled excited state ²H_{11/2} and ⁴S_{3/2} levels of Er³⁺ in both phases of phosphors were recorded at temperatures ranging from room temperature to 573 K under 980 nm infrared laser excitation. The time scanning upconversion spectra were investigated in detail to reveal the thermal effect induced by laser irradiation and the luminescent thermal stability of the two phases NaYF₄ polycrystals. Comparison of the upconversion luminescence and the sensitivity between the two phases NaYF₄ polycrystals indicated that β -phase NaYF₄ won much stronger luminescent intensity, better luminescent thermal stability, and higher temperature sensitivity.

© 2015 Elsevier B.V. All rights reserved.

1. Introduction

During the past decades, rare-earth (RE) doped upconversion (UC) micro-/nano-crystals have attracted considerable attention thanks to their potential applications in many fields, such as three dimension flat-panel displays, drug delivery, biomedical imaging and sensitive bio-probes [1–6], etc. Among these UC materials, RE fluorides with the general molecular formula NaREF₄ are often employed as host materials to date owing to their low phonon energy, high chemical stability, and high refractive index [7]. There are two typical crystal structure of NaREF₄, namely, α -phase and β -phase. In these fluorides, β -phase NaYF₄ has been regarded as the most efficient UC host material for the UC luminescence (UCL) and received growing interest in recent years [1,3,8–10].

Er³⁺ has been widely recognized as an ideal doping center for UCL in considering its favorable intermediate levels of ⁴I_{9/2}, ⁴I_{11/2} and ⁴I_{13/2} which can be populated easily by commercial infrared or near infrared laser diodes (LDs), or by accepting the excitation energy transferred from other sensitizer, for example Yb³⁺. In addition, Er³⁺ has two neighbored levels of ²H_{11/2} and ⁴S_{3/2} whose populations meet the Boltzmann distribution law, thus the

temperature of Er³⁺-doped materials can be obtained by processing the fluorescence intensity ratio (FIR) of the two green emissions from these two excited states ²H_{11/2} and ⁴S_{3/2} to the ground state ⁴I_{15/2} level. Recently, the UCL and temperature sensing properties of Er³⁺/Yb³⁺ co-doped NaYF₄ has been extensively studied [11,12]. However, there is no such knowledge about which one amongst these two phases NaYF₄:Er³⁺/Yb³⁺ crystals is better for the application in temperature sensing and which one has a better stability of temperature dependent UCL.

Up to now, many preparative techniques including co-precipitation, hydrothermal reaction, and high temperature thermal decomposition, etc. have been employed to synthesize RE-doped NaYF₄ material [8–10,13–18]. Microwave hydrothermal is an improved hydrothermal synthesis approach and increasingly applied in the preparation of micro-/nano-sized materials [17–19]. Compared with the conventional hydrothermal method, microwave-assisted hydrothermal has some obvious advantages, such as higher heating rates, shorter reaction time, better controlling of the reaction parameters, higher yields, and excellent reproducibility due to its unique thermal effect induced by microwave heating [19].

In this paper, α - and β -phase NaYF₄:Yb³⁺/Er³⁺ nano-/micro-crystals were successfully synthesized by a microwave-assisted hydrothermal method. Both the UCL and temperature sensing

* Corresponding authors. Tel./fax: +86 41184728909.

E-mail addresses: lixp@dlmu.edu.cn (X. Li), chenmbj@sohu.com (B. Chen).

properties of the nano-/micro-crystals were systematically studied.

2. Experimental

2.1. Synthesis of $\text{NaYF}_4: 20\% \text{Yb}^{3+}/2\% \text{Er}^{3+}$ nano-/micro-crystals.

All the lanthanide oxides including spectroscopically pure Y_2O_3 (99.99%), Er_2O_3 (99.99%), and Yb_2O_3 (99.99%) were purchased from Shanghai Second Chemical Reagent Factory (China). Other chemicals including trisodium citrate (Na_3Cit) and ethylenediamine tetraacetic acid (EDTA) were purchased from Tianjin Reagent Chemicals Co. Ltd., (China). All chemical reagents except for the RE_2O_3 were analytical grade and were used without any further purification. In the synthesis process, the lanthanide chlorates $\text{LnCl}_3 \cdot 6\text{H}_2\text{O}$ ($\text{Ln} = \text{Y}$, Yb , and Er) were used instead of the corresponding Ln_2O_3 . These $\text{LnCl}_3 \cdot 6\text{H}_2\text{O}$ were obtained by dissolving the corresponding Ln_2O_3 in a proper amount of hydrochloric acid and re-crystallized three times.

A commercial microwave synthesizer system (Biotage Initiator, Sweden) was used to synthesize the $\text{NaYF}_4: \text{Yb}^{3+}/\text{Er}^{3+}$ phosphors. Both EDTA and Na_3Cit were used as additives for controlling the crystal structure of the products [20,21]. In a typical process, 1.58 mmol EDTA or 3.16 mmol Na_3Cit was separately added into deionized water under vigorous stirring until completely dissolved to form a homogeneous solution. Then 2.46 mL $\text{YCl}_3 \cdot 6\text{H}_2\text{O}$ (1.0 M), 6.32 mL $\text{YbCl}_3 \cdot 6\text{H}_2\text{O}$ (0.40 M) and 6.32 mL $\text{ErCl}_3 \cdot 6\text{H}_2\text{O}$ (0.04 M) were dropped slowly into the above solution to form the precursor. After vigorously stirring for 15 min and sonicating for 30 min, another mixture solution of 20 mL ethanol glycol (1.0 M) and 38 mL NaF (1.0 M) was added into the precursor solution. Subsequently, the as-obtained mixture solution was transferred to a 30 mL silica glass vessel sealed with a special cap provided by the manufacturer. Then the vessel was put into the microwave synthesizer and irradiated for 30 min at 160 °C. After microwave heating, the vessel was naturally cooled to room temperature. Then, the products were washed with deionized water several times and dried at 80 °C.

2.2. Sample characterization

The X-ray diffraction (XRD) patterns of the as-prepared samples were collected using a Shimadzu XRD-6000 diffractometer with $\text{CuK}\alpha$ radiation source ($\lambda = 0.15406 \text{ nm}$). The morphologies were observed on a Hitachi S-4800 field emission scanning electron microscope (FE-SEM). The UCL spectra were obtained by a Hitachi F-4600 fluorescent spectrometer pumped by a power-adjustable 980 nm fiber laser with maximum output power of 2 W. The temperatures of the studied samples were controlled by a sample temperature controlling system DMU-TC 450 assembled in our lab. With this system the sample temperature can be well adjusted and controlled in the temperature region from room temperature to 450 °C.

3. Results and discussion

3.1. Crystal structure and morphology characterization

In order to identify the crystal structure of the as-obtained products, XRD measurements were performed. Fig. 1 shows the XRD patterns of the samples synthesized using different chelators of EDTA (Fig. 1a) and Na_3Cit (Fig. 1c) together with the patterns plotted by using the data reported in JCPDS Card nos. 77-2042 (Fig. 1b) and 28-1192 (Fig. 1d) corresponding to the α - and β -phase

NaYF_4 powders. It can be found that most of the diffraction peaks of the sample synthesized using EDTA as chelator are coincidence with the one plotted in Fig. 1b corresponding to the α -phase NaYF_4 . Meanwhile, some very weak extra diffraction peaks marked by asterisks (*) can also be observed in Fig. 1a, which corresponds to another α -phase NaYF_4 (JCPDS no. 39-0724). While all the diffraction peaks of the sample synthesized using Na_3Cit as chelator are in agreement well with those displayed in Fig. 1d corresponding to the β -phase NaYF_4 . And no obviously additional diffraction peaks from other impurities can be detected in Fig. 1c. These results indicate that the introduction of Er^{3+} and Yb^{3+} does not obviously affect the crystal structure of the host NaYF_4 , and that different phases of NaYF_4 crystals can be obtained by adopting different chelators via the microwave hydrothermal reaction.

Fig. 2 shows the FE-SEM images of the samples prepared under different reaction parameter conditions. When using EDTA as chelator, the sample exhibits irregular morphology and composes of small spherical nanoparticles corresponding to the α -phase NaYF_4 , as shown in Fig. 2a. These small spherical nanoparticles present a rough surface and the mean diameter of the particles is

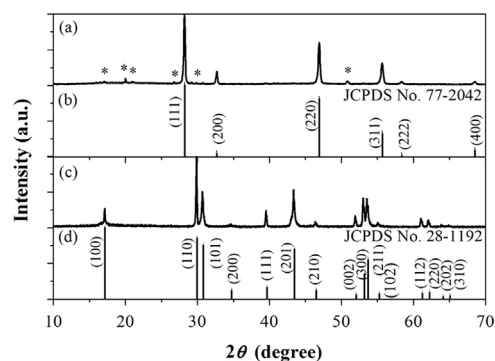


Fig. 1. XRD patterns of the samples synthesized using different chelators: EDTA (a) and Na_3Cit (c), and the standard XRD patterns appeared in JCPDS Card nos. 77-2042 (b) and 28-1192 (d).

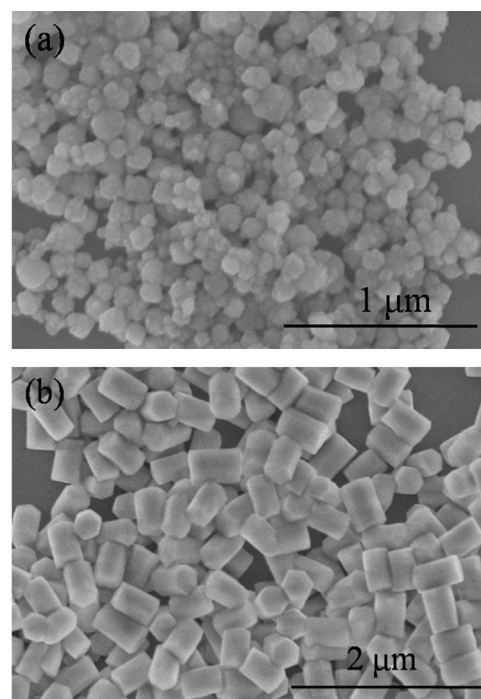


Fig. 2. FE-SEM images for samples synthesized using EDTA (a) and Na_3Cit (b) as chelators.

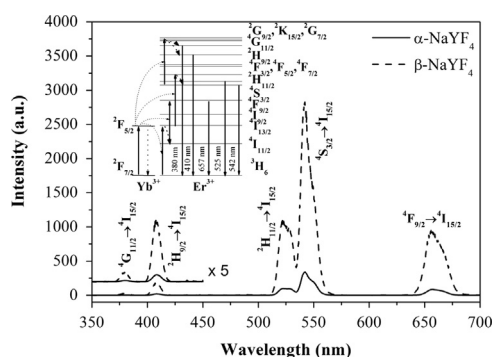


Fig. 3. UCL spectra for α - and β -NaYF₄:20%Yb³⁺/2%Er³⁺ excited by 980 nm LD with pump power density of 2 W/cm². A. Inset: energy level diagram of Er³⁺ and Yb³⁺ in NaYF₄ and possible upconversion mechanism under 980 nm LD excitation.

about 75 nm. A large quantity of regular short hexagonal rods corresponding to the β -phase NaYF₄ is obtained when using Na₃Cit as chelator. The mean size of the short hexagonal rods is about 200 nm in diameter and 400 nm in length.

3.2. Upconversion luminescence properties

The UCL spectra for α - and β -NaYF₄:Yb³⁺/Er³⁺ nano-/micro-crystals were recorded upon excitation of 980 nm LD working at various pump power densities (from 0.6 W/cm² to 5.9 W/cm²). It is worthwhile pointing out that all the experimental conditions were kept identical throughout the spectrum measurements. Firstly, the area of the laser beam spot was kept constant. Secondly, each UC spectrum was measured just at the beginning of the laser irradiated at the sample. Finally, the time interval between each two adjacent measurements was set to be long enough to ensure that the sample has been cooled down to room temperature before each measurement. In this case, the thermal effect induced by laser irradiation can be avoided expectedly [19]. Fig. 3 shows the UCL spectra of both α - and β -NaYF₄:20%Yb³⁺/2%Er³⁺ nano-/micro-crystals pumped with pump power density of 2 W/cm² as representatives. Both UCL spectra appear intense green and red emissions originating from (²H_{11/2}, ⁴S_{3/2}) and ⁴F_{9/2} excited state levels of Er³⁺. In addition, weak purple and blue UC emissions centered at around 379 and 408 nm can also be found (the corresponding spectra have been magnified by a factor of 5 for clear observation), which can be assigned to the transitions ⁴G_{11/2} → ⁴I_{15/2} and ²H_{9/2} → ⁴I_{15/2} of Er³⁺. The inset of Fig. 3 shows the energy level diagram of Er³⁺ and Yb³⁺ in NaYF₄ and possible upconversion mechanism under 980 nm LD excitation. As shown in Fig. 3, the β -phase sample exhibits stronger UCL intensity than that of the α -phase one, which can be ascribed to the much more effective energy transfer in β -phase sample, additionally, the decrease of the nonradiative relaxation channel induced by the larger size of hexagonal rods than that of the spherical nanoparticles [7,22,23].

It is well known that the sample temperature can be elevated when the sample is exposed to laser, especially under the irradiation of high power density laser [24]. In order to understand the thermal effect of laser irradiation on the UCL, the time scanning UCL spectra for both α - and β -phase samples with varied pump power densities of 980 nm laser were measured by monitoring the strongest green UC emission (545 nm) at room temperature. During each measurement, the samples were continuously irradiated by 980 nm LD. Apart from that, each measurement was started when the sample temperature was room temperature.

Fig. 4 shows the normalized time scanning UCL spectra measured under 980 nm excitation by monitoring 545 nm emission for

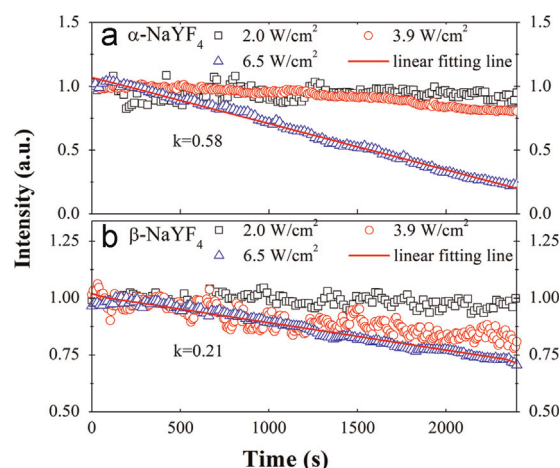


Fig. 4. Normalized time scanning UCL spectra for both α - and β -phase samples measured at various pump power densities under 980 nm LD excitation. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

both α - and β -phase samples. In Fig. 4 the X-axis presents the irradiation time of 980 nm laser, and Y-axis is the normalized UCL intensity. It can be observed that when the pump power density is 2 W/cm², the green UCL intensities for both α - and β -phase samples almost not change with increasing laser irradiation time. The UCL intensities for both samples decrease slightly with irradiation time in the case of a higher pump power density of 3.9 W/cm². When the pump power density increases to 6.5 W/cm², the UCL intensities for both samples decrease dramatically with increasing laser irradiation time, which reflects a significant quenching of the UCL. This may be caused by the increase of the multiphonon cascade nonradiative relaxation of UC emission level for the elevated temperature of the samples induced by the continuous laser irradiation [26].

It should be noted that when the pump power density is 6.5 W/cm², the UCL intensities for both samples still decrease after continued irradiation of 40 min and exhibit linear decrease relation with irradiation time. The slope of the linear fitting line can reflect the change rate of the UCL intensity, meaning the change of UCL intensity per unit time. The higher the slope value the higher the changing rate of the UCL intensity, which implies that the sample has much poorer luminescence thermal stability. The linear fitting results for the two sets of experimental data are plotted in Fig. 4 marked by the solid red lines. The slope values are 0.58 and 0.21 for α - and β -phase samples, respectively. That is to say, the change rate of the UCL intensity of α -phase sample is far larger than that of the β -phase one. Therefore, it can be deduced that the UCL of Er³⁺/Yb³⁺ co-doped β -phase sample has much better luminescent thermal stability than that of α -phase one.

3.3. Optical temperature sensing properties

Optical temperature sensing has been extensively investigated on the Er³⁺-doped system due to the narrow energy gap between the ²H_{11/2} and ⁴S_{3/2} levels of Er³⁺ [11,12]. These two green emission levels are always in a thermal equilibrium at any temperatures due to the thermally coupled regime. The populations of ²H_{11/2} and ⁴S_{3/2} fulfill the Boltzmann distribution law. The relationship between the FIR (*R*) of these two green emissions and temperature can be expressed as below [25]

$$R = \frac{I_H}{I_S} = C \exp\left(-\frac{\Delta E}{kT}\right) \quad (1)$$

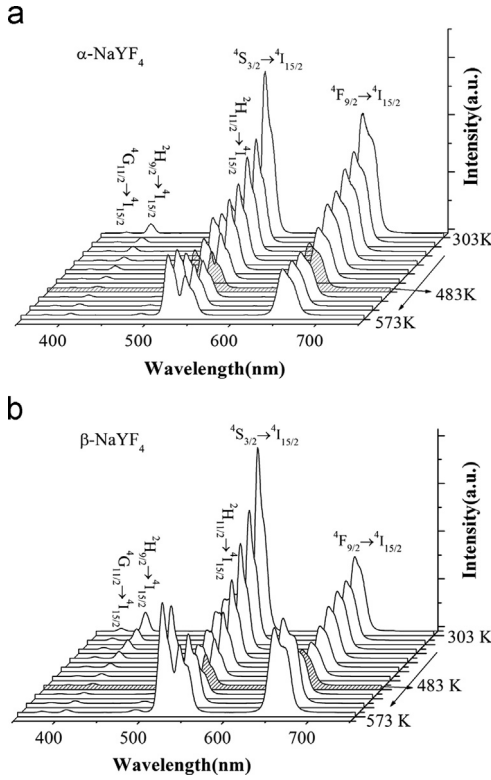


Fig. 5. Temperature dependent UCL spectra for α - and β -NaYF₄: 20%Yb³⁺/2%Er³⁺ excited by 980 nm laser.

where I_H and I_S are the integrated intensities of the emissions for $^2H_{11/2} \rightarrow ^4I_{15/2}$ and $^4S_{3/2} \rightarrow ^4I_{15/2}$ transitions of Er³⁺, respectively. C is a constant which is independent from the sample temperature but dependent on the spontaneous emission rate, degeneracy and photon energy. ΔE represents the energy gap between the $^2H_{11/2}$ and $^4S_{3/2}$ levels, k is the Boltzmann constant, and T is the absolute temperature. From Eq. (1), it can be easily found that the value of R is only associated with the variable T , thus the target temperature of the studied system can be determined on the basis of the measurement of the intensity ratio R .

From the discussion in above section, it can be concluded that the thermal effect induced by laser irradiation can be neglected when the pump power density as low as 2 W/cm². Therefore, in the following temperature sensing measurements, 2 W/cm² was chosen as the pump power density of LD. In this way, the intensity ratio R can truly reflect the temperature change of the sample. Fig. 5 shows the temperature dependent UCL spectra for both α - and β -phase samples excited by 980 nm LD in the temperature region of 303–573 K, which is lower than the phase transition temperature of NaYF₄ host [26,27]. It can be seen that with increasing temperature, the intensities of both the emissions from $^2H_{11/2} \rightarrow ^4I_{15/2}$ and $^4S_{3/2} \rightarrow ^4I_{15/2}$ transitions have the same varied trends. Both of them decrease first and reach the minimum value at around 483 K, and then increase with further increase of temperature. This anomalous temperature dependent UCL behavior has been systematically studied in our other published paper [28]. The initial decline of the UCL belongs to a typical thermal quenching process and can be attributed to the poor crystallinity and the enhancement of multi-phonon relaxation rates at high temperature. When the sample temperature reaches a sufficiently high temperature (such as 513 K), the lattice of the host can be reorganized, at the same time, the adsorbed water on the sample and the residual organic components in the synthesized process can be desorbed or decomposed, resulting in the increase of UCL at high temperature.

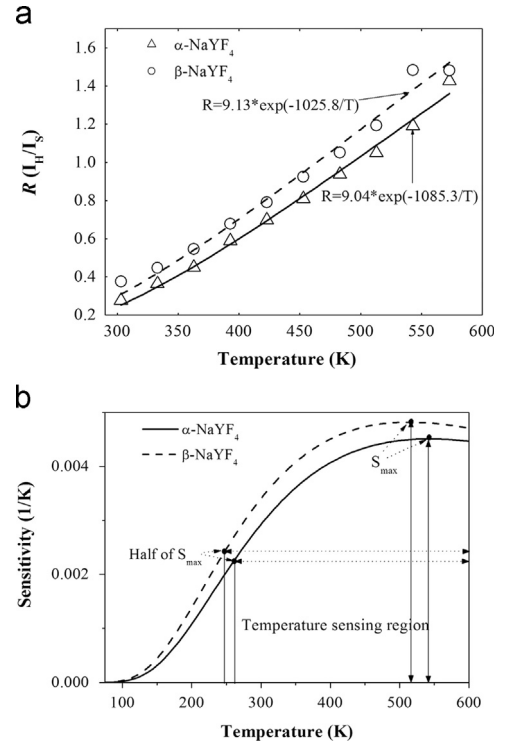


Fig. 6. (a) Relationship between the intensity ratio R and the temperature, (b) dependence of the sensitivity on the temperature.

However, the intensity ratios R continue to increase in both samples with increasing temperature, which can be found in Fig. 6. The open symbols in Fig. 6 show the dependence of the intensity ratio R on the temperature in the temperature region of 303–573 K for both the samples. Eq. (1) was used to fit the experimental data, and the fitted results are shown in Fig. 6a marked as the solid and dash curves. From the fitting processes, the values of the free parameters C and $\Delta E/k$ in Eq. (1) were confirmed to be 9.04 and 1085.3 K for α -phase sample, 9.13 and 1025.8 K for β -phase sample.

The sensitivity of Er³⁺-based temperature sensor is defined as the rate of R change toward temperature change, and can be mathematically expressed as [29]

$$S(T) = \frac{dR(T)}{dT} = C \exp\left(-\frac{\Delta E}{kT}\right) \left(\frac{\Delta E}{kT^2}\right) \quad (2)$$

By taking the above-derived values of C and $\Delta E/k$ into Eq. (2), the sensitivity functions $S(T)$ versus T for both samples can be derived and are plotted in Fig. 6b in the temperature region of 75–600 K. It can be seen that the sensitivity S of each sample changes greatly with temperature and the maximum sensitivities of $45.4 \times 10^{-4} \text{ K}^{-1}$ and $48.4 \times 10^{-4} \text{ K}^{-1}$ are achieved at around 541 K and 515 K for α - and β -phase samples, respectively. Moreover, we can also find that the sensitivity of β -phase sample is a little higher than that of the α -phase one in the whole measured temperature range, indicating a better temperature sensing property of β -phase sample. Generally, we consider that when the sensitivity S is higher than half of its maximum value $S_{\max}$, the temperature sensing result is reliable. As can be seen from Fig. 6b, the two phase's samples both have large temperature sensing region locating between 247 K (or 262 K) and 600 K for β - (or α -) phase sample, respectively. However, it should also be noted that the β -phase sample exhibits much intenser UC emission at the same excitation density, which is beneficial to the practical applications.

4. Conclusion

In conclusion, α - and β -NaYF₄:Yb³⁺/Er³⁺ micro-/nano-crystals were successfully prepared via a microwave-assisted hydrothermal route. The final products of α - and β -phase samples are present as spherical particles with mean diameter about 75 nm and short hexagonal rods with mean size of 200 nm in diameter and 400 nm in length, respectively. The UCL and temperature sensing properties of both samples were investigated and compared. The results showed that intense green, red and weak blue, purple UC emissions were found in the UCL spectra of both samples and β -phase sample had stronger intensity. From time scanning UCL spectra, it is found that the change rate of the UCL intensity depended on the pump power density of LD. When the pump power density as low as 2 W/cm², the UCL intensity almost remained constant with increasing irradiation time, but decreased dramatically under a higher pump power density of 6.5 W/cm². Moreover, both samples exhibit good temperature sensing properties and β -phase sample had better sensing property. The maximum sensitivities were confirmed to be 45.4×10^{-4} and $48.4 \times 10^{-4} \text{ K}^{-1}$ at around 541 K and 515 K for α - and β -phase samples, respectively.

Acknowledgments

This work is financially supported by the National Natural Science Foundation of China (Grant nos. 11104023, 11374044 and 11104024), the Fundamental Research Funds for the Central Universities (3132014087 and 3132014327).

References

- [1] Q.B. Xiao, Y.T. Ji, Z.H. Xiao, Y. Zhang, H.Z. Lin, Q.B. Wang, Novel multifunctional NaYF₄:Er³⁺,Yb³⁺/PEGDA hybrid microspheres: NIR-light-activated photopolymerization and drug delivery, *Chem. Commun.* 49 (2013) 1527–1529.
- [2] E. Downing, L. Hesselink, J. Ralston, R. Macfarlane, A three-color, solid-state, three-dimensional display, *Science* 273 (1996) 1185–1189.
- [3] G.S. Yi, H.C. Lu, S.Y. Zhao, Y. Ge, W.J. Yang, D.P. Chen, L.H. Guo, Synthesis, characterization, and biological application of size-controlled nanocrystalline NaYF₄:Yb,Er infrared-to-visible up-conversion phosphors, *Nano Lett.* 4 (2004) 2191–2196.
- [4] T. Sandrock, H. Scheife, E. Heumann, G. Huber, High-power continuous-wave upconversion fiber laser at room temperature, *Opt. Lett.* 22 (1997) 808–810.
- [5] L. Rao, W. Lu, H.B. Wang, Z.G. Yi, S.J. Zeng, Z. Li, Sub-10 nm lanthanide doped BaLuF₅ nanocrystals: shape controllable synthesis, tunable multicolor emission and enhanced near-infrared upconversion luminescence, *Mater. Res. Bull.* 64 (2015) 27–32.
- [6] S.J. Zeng, J.J. Xiao, Q.B. Yang, J.H. Hao, Bi-functional NaLuF₄:Gd³⁺/Yb³⁺/Tm³⁺ nanocrystals: structure controlled synthesis, near-infrared upconversion emission and tunable magnetic properties, *J. Mater. Chem.* 22 (2012) 9870–9874.
- [7] K.W. Kramer, D. Biner, G. Frei, H.U. Gudel, M.P. Hehlen, S.R. Luthi, Hexagonal sodium yttrium fluoride based green and blue emitting upconversion phosphors, *Chem. Mater.* 16 (2004) 1244–1251.
- [8] S.Y. Yu, X.C. Gao, H. Jing, J. Zhao, H.Q. Su, A synthesis and up-conversion photoluminescence study of hexagonal phase NaYF₄:Yb,Er nanoparticles, *CrystEngComm* 15 (2013) 10100–10106.
- [9] M.Y. Ding, C.H. Lu, L.H. Cao, Y.R. Ni, Z.Z. Xu, Controllable synthesis, formation mechanism and upconversion luminescence of β -NaYF₄:Yb³⁺/Er³⁺ micro-crystals by hydrothermal process, *CrystEngComm* 15 (2013) 8366–8373.
- [10] K.Z. Zheng, D. Zhao, D.S. Zhang, N. Liu, W.P. Qin, Temperature-dependent six-photon upconversion fluorescence of Er³⁺, *J. Fluorine Chem.* 132 (2011) 5–8.
- [11] B.S. Cao, Y.Y. He, L. Zhang, B. Dong, Upconversion properties of Er³⁺-Yb³⁺:NaYF₄ phosphors with a wide range of Yb³⁺ concentration, *J. Lumin.* 135 (2013) 128–132.
- [12] K.Y. Wu, J.B. Cui, X.X. Kong, Y.J. Wang, Temperature dependent upconversion luminescence of Yb/Er codoped NaYF₄ nanocrystals, *J. Appl. Phys.* 110 (2011) 053510.
- [13] J.C. Boyer, F. Vetrone, L.A. Cuccia, J.A. Capobianco, Synthesis of colloidal upconverting NaYF₄ nanocrystals doped with Er³⁺,Yb³⁺ and Tm³⁺,Yb³⁺ via thermal decomposition of lanthanide, *J. Am. Chem. Soc.* 128 (2006) 7444–7445.
- [14] N.J. Johnson, N.M. Sangeetha, J.C. Boyer, F.C. van Veggel, Facile ligand-exchange with polyvinylpyrrolidone and subsequent silica coating of hydrophobic upconverting β -NaYF₄:Yb³⁺/Er³⁺ nanoparticles, *Nanoscale* 2 (2010) 771–777.
- [15] M. Wang, C.C. Mi, S. Wang, F. Li, J.L. Liu, S.K. Xu, Synthesis and characterization of NaYF₄:Yb, Er upconversion fluorescent nanoparticles via a co-precipitation method, *Spectrosc. Spect. Anal.* 29 (2009) 3327–3331.
- [16] F. Wang, Y. Han, C.S. Lim, Y.H. Lu, J. Wang, J. Xu, H.Y. Chen, C. Zhang, M.H. Hong, X.G. Liu, Simultaneous phase and size control of upconversion nanocrystals through lanthanide doping, *Nature* 463 (2010) 1061–1065.
- [17] X. Chen, W.J. Wang, X.Y. Chen, J.H. Bi, L. Wu, Z.H. Li, X.Z. Fu, Microwave hydrothermal synthesis and upconversion properties of NaYF₄:Yb³⁺,Tm³⁺ with microtube morphology, *Mater. Lett.* 63 (2009) 1023–1026.
- [18] L. Zhang, Y.J. Zhu, Microwave hydrothermal synthesis of hexagonal NaYF₄ and Yb³⁺,Er³⁺ doped NaYF₄ microtube, *J. Inorg. Mater.* 24 (2009) 553–558.
- [19] H. Zheng, B.J. Chen, H.Q. Yu, J.S. Zhang, J.S. Sun, X.P. Li, M. Sun, B.N. Tian, S.B. Fu, H. Zhong, B. Dong, R.N. Hua, H.P. Xia, Microwave-assisted hydrothermal synthesis and temperature sensing application of Er³⁺/Yb³⁺ doped NaY(WO₄)₂ microstructures, *J. Colloid Interface Sci.* 420 (2014) 27–34.
- [20] Y. Tian, R.N. Hua, B.J. Chen, N.S. Yu, W. Zhang, L.Y. Na, Lanthanide dopant-induced phase transition and luminescence enhancement of EuF₃ nanocrystals, *CrystEngComm* 14 (2012) 8110–8116.
- [21] H.L. Qiu, G.Y. Chen, L. Sun, S.W. Hao, G. Han, C.H. Yang, Ethylenediaminetetraacetic acid (EDTA)-controlled synthesis of multicolor lanthanide doped BaYF₅ upconversion nanocrystals, *J. Mater. Chem.* 21 (2011) 17202–17208.
- [22] Y.J. Sun, Y. Chen, L.J. Tian, Y. Yu, X.G. Kong, J.W. Zhao, H. Zhang, Controlled synthesis and morphology dependent upconversion luminescence of NaYF₄:Yb,Er nanocrystals, *Nanotechnology* 18 (2007) 275609.
- [23] Y. Tian, R.N. Hua, N.S. Yu, W. Zhang, L.Y. Na, B.J. Chen, Morphology and phase evolution of Yb³⁺/Er³⁺ co-doped NaYF₄ microtubes prepared with YF₃ submicrospindles as precursor, *J. Alloy. Compd.* 509 (2011) 9924–9929.
- [24] S.Y. Xiang, B.J. Chen, J.S. Zhang, X.P. Li, J.S. Sun, H. Zheng, Z.L. Wu, H. Zhong, H.Q. Yu, H.P. Xia, Microwave-assisted hydrothermal synthesis and laser-induced optical heating effect of NaY(WO₄)₂:Tm³⁺/Yb³⁺ microstructures, *Opt. Mater. Express* 4 (2014) 1966–1980.
- [25] J.J. Li, J.S. Sun, J.T. Liu, X.P. Li, J.S. Zhang, Y. Tian, S.B. Fu, L.H. Cheng, H.Y. Zhong, H.P. Xia, B.J. Chen, Pumping-route-dependent concentration quenching and temperature effect of green up- and down-conversion luminescence in Er³⁺/Yb³⁺ co-doped Gd₂(WO₄)₃ phosphors, *Mater. Res. Bull.* 48 (2013) 2159–2165.
- [26] H.X. Mai, Y.W. Zhang, R. Si, Z.G. Yan, L.D. Sun, L.P. You, C.H. Yan, High-quality sodium rare-earth fluoride nanocrystals: controlled synthesis and optical properties, *J. Am. Chem. Soc.* 128 (2006) 6426–6436.
- [27] J.N. Shan, Y.G. Ju, Controlled synthesis of lanthanide-doped NaYF₄ upconversion nanocrystals via ligand induced crystal phase transition and silica coating, *Appl. Phys. Lett.* 91 (2007) 123103.
- [28] L.L. Tong, X.P. Li, R.N. Hua, T.X. Peng, Y.Z. Wang, X.Z. Zhang, B.J. Chen, Anomalous temperature-dependent upconversion luminescence of β -NaYF₄:Yb³⁺/Er³⁺ nanocrystals synthesized by a microwave-assisted hydrothermal method, *J. Nanosci. Nanotechnol.* 15 (2015) 1–6.
- [29] F. Verone, R. Naccache, A. Zamarron, A.J. Fuente, F. Sanz-Rodriguez, L.M. Maestro, E.M. Rodriguez, D. Jaque, J.G. Sole, J.A. Capobianco, Temperature sensing using fluorescent nanothermometers, *ACS Nano* 4 (2010) 3254–3258.